TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of Water Hardness and Removal of Hardness

Why dissolved calcium and magnesium salts are the first enemy of wet processing, how the EDTA titration measures them as $CaCO_3$, and how boiling, the lime–soda process, ion exchange and sequestration remove them.

TE-2206 · TEXTILE CHEMISTRY & COLORATION-I SESSIONAL · EXPERIMENT 1 · LEVEL 2 TERM 2

1. Aim and theory

Aim: to determine the total hardness of a given water sample by complexometric titration with EDTA, to separate it into temporary and permanent hardness, and to study the methods by which hardness is removed. Water is the largest single raw material of a dyehouse — a scouring or dyeing bath may carry 20–50 litres of water per kilogram of goods — so the quality of the water decides the quality of everything done in it.

What hardness is

Hardness is the property water acquires from dissolved salts of **calcium and magnesium** (with iron, manganese and strontium contributing in practice). Rain picks up the salts as it passes over limestone, chalk, dolomite and gypsum beds. The salts fall into two families whose behaviour is completely different:

TYPE	SALTS RESPONSIBLE	BEHAVIOUR
Temporary (carbonate) hardness	$Ca(HCO_3)_2$ and $Mg(HCO_3)_2$ — bicarbonates, held in solution only by dissolved CO_2	Removed by boiling: the bicarbonate decomposes and the carbonate precipitates as scale or sludge
Permanent (non-carbonate) hardness	$CaCl_2$, $MgCl_2$, $CaSO_4$, $MgSO_4$ — chlorides and sulphates	Unaffected by boiling; needs chemical treatment or ion exchange

$$Ca(HCO_3)_2 \xrightarrow{\text{heat}} CaCO_3\downarrow + CO_2\uparrow + H_2O$$

Unit. Because a water contains a mixture of salts, hardness is always reported as the equivalent amount of $CaCO_3$, in **mg/L (numerically equal to ppm)**. Any salt is converted by its equivalent weight, since one equivalent of any hardness salt does the same chemical work as one equivalent (50 g) of $CaCO_3$:

$$\text{Hardness as } CaCO_3 = (\text{mg/L of salt}) \times 50 / (\text{equivalent weight of the salt})$$

The EDTA method

EDTA (ethylenediaminetetraacetic acid, used as its disodium salt, written H_2Y^{2-}) is a **complexing (chelating) agent**: one EDTA ion wraps around one Ca^{2+} or Mg^{2+} ion and holds it in a soluble, colourless, very stable complex. The combining ratio is exactly **1 : 1**, whatever the metal:

$$M^{2+} + H_2Y^{2-} \rightarrow MY^{2-} + 2\,H^{+} \quad (M = Ca \text{ or } Mg)$$

The indicator is **Eriochrome Black T (EBT)**. In the sample at pH 10 it first binds a little of the Mg^{2+}/Ca^{2+} and shows **wine red**. As EDTA is run in, it takes the free metal ions first and finally strips the metal from the indicator itself; the freed indicator at pH 10 is **blue**. The end point is the change from wine red to a pure blue with no reddish tinge. The pH is held at 10.0 by an **ammonia–ammonium chloride buffer**: below pH 10 the indicator colour change is indistinct, and much above it magnesium hydroxide begins to precipitate. EBT responds best when a little Mg^{2+} is present, which most natural waters supply.

Why hardness matters in wet processing

- **Soap is wasted and scum is formed.** Calcium and magnesium precipitate soap as an insoluble curd that deposits on the goods and the machine:

$$2\,C_{17}H_{35}COONa + Ca^{2+} \rightarrow (C_{17}H_{35}COO)_2Ca\downarrow + 2\,Na^{+}$$

- **Deposits on the fabric:** lime soaps and precipitated $CaCO_3$ give a harsh handle, greyish cast and poor absorbency — the exact faults preparation exists to remove (Experiment 3 tests for them).
- **Uneven dyeing:** Ca^{2+}/Mg^{2+} form lakes and complexes with many anionic dyes and with dyeing auxiliaries, shifting shade and causing unlevel, specky results; metal-sensitive dyes can change hue entirely.
- **Scale and machinery damage:** temporary hardness precipitates in boilers, heat exchangers and J-boxes, wasting fuel and eventually blocking circulation.

A textile wet-processing unit therefore works to a water standard, not to whatever the supply brings: total hardness below about **50 mg/L as $CaCO_3$** is the usual target for processing, and still lower for critical dyeing.

Removing hardness

(a) Boiling removes temporary hardness only (reaction above). It is exact chemistry but absurd economics at factory scale, and it leaves every permanent salt behind; its laboratory use is to *measure* permanent hardness, as done in this experiment.

(b) Lime–soda process. Slaked lime, $Ca(OH)_2$, removes temporary hardness and all magnesium hardness; soda ash, Na_2CO_3, removes permanent calcium hardness. The products are insoluble $CaCO_3$ and $Mg(OH)_2$, settled and filtered out:

$$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2\,CaCO_3\downarrow + 2\,H_2O$$

$$MgCl_2 + Ca(OH)_2 \rightarrow Mg(OH)_2\downarrow + CaCl_2 \quad \text{then} \quad CaCl_2 + Na_2CO_3 \rightarrow CaCO_3\downarrow + 2\,NaCl$$

$$CaSO_4 + Na_2CO_3 \rightarrow CaCO_3\downarrow + Na_2SO_4$$

Lime also destroys free CO_2 ($CO_2 + Ca(OH)_2 \rightarrow CaCO_3\downarrow + H_2O$), which otherwise consumes chemicals later. The process is cheap for large volumes but needs correct dosing — excess lime or soda is itself a contaminant — and it produces sludge.

(c) Ion exchange (zeolite/permutit and synthetic resins). The water passes through a bed of sodium-form exchanger, Z, which swaps its Na^+ for Ca^{2+} and Mg^{2+}:

$$Ca^{2+} + Na_2Z \rightarrow CaZ + 2\,Na^+ \quad ; \quad \text{regeneration: } CaZ + 2\,NaCl \rightarrow Na_2Z + CaCl_2$$

When the bed is exhausted it is regenerated with brine. Ion exchange removes *both* hardness types to near zero and is the standard factory softener; a two-bed (H^+/OH^- resin) plant goes further to fully demineralised water. Note the exchanged water holds sodium salts instead — soft, but not salt-free.

(d) Sequestering (chelating) agents. Polyphosphates (Calgon), EDTA, NTA and citrates do not remove Ca^{2+}/Mg^{2+} at all — they lock them in soluble complexes so they cannot precipitate soap or disturb dyeing. This is the remedy used *inside the bath* (in scouring and dyeing liquors) when softening plant water is impossible or as a safety net. Lime–soda and ion exchange soften the *supply*; sequestration protects the *process*.

2. Apparatus and procedure

- Burette (50 mL), pipette (50 mL and 25 mL), conical flasks, measuring cylinder, beaker, funnel, filter paper, white tile, tripod and burner (for the boiling step).
- Standard EDTA solution, 0.0100 M; buffer solution pH 10 ($NH_4Cl + NH_4OH$); Eriochrome Black T indicator (solid mixture with NaCl, or alcoholic solution); the water sample.

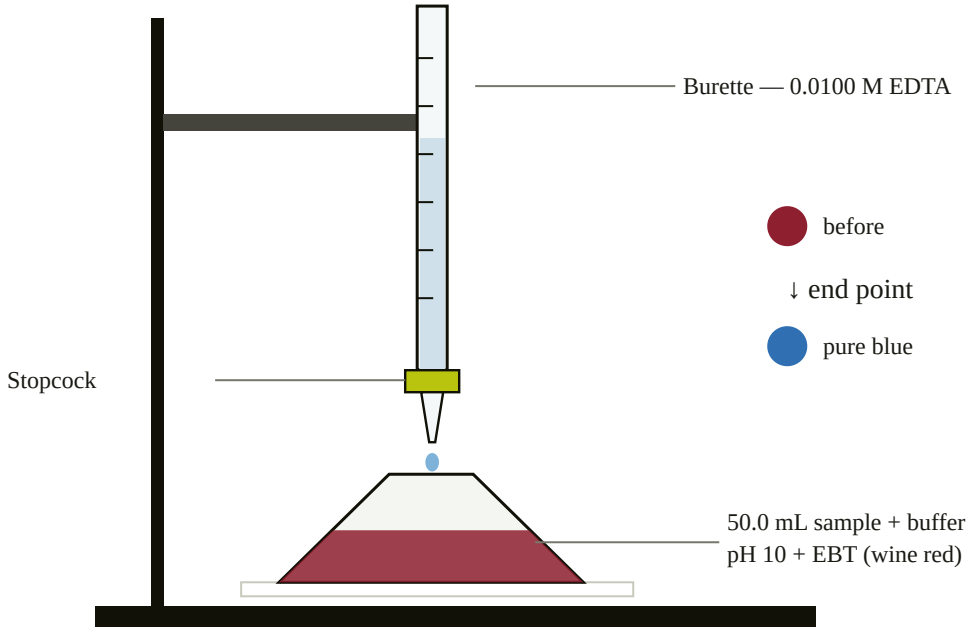


Fig. 1 — EDTA titration set-up. The flask stands on a white tile so the end point — wine red to pure blue — is seen without doubt.

Total hardness

1. Rinse the burette with the EDTA solution, fill it, and record the initial reading with the meniscus at eye level.
2. Pipette 50.0 mL of the water sample into a conical flask. Add about 2 mL of the pH 10 buffer and a pinch of the EBT mixture (or 2–3 drops of indicator solution). The solution turns wine red.
3. Titrate with EDTA, swirling continuously, slowing to single drops as the red begins to fade. The end point is the first permanent **pure blue** colour with no reddish tinge.
4. Repeat until two titres agree within 0.1 mL and average them.

Permanent hardness — and temporary by difference

1. Boil 250 mL of the sample gently for about 30 minutes; the temporary-hardness carbonates precipitate. Filter while hot, wash the precipitate with a little distilled water, cool the filtrate and make it back up to 250 mL with distilled water.
2. Titrate 50.0 mL of this treated water exactly as above. This titre measures **permanent hardness**.
3. **Temporary hardness = total hardness − permanent hardness.**

3. Worked calculation (model data)

DETERMINATION	SAMPLE VOLUME	TITRES OF 0.0100 M EDTA (ML)

Total hardness	50.0 mL	12.6, 12.5, 12.5 → mean 12.5
Permanent hardness (after boiling)	50.0 mL	8.1, 8.0 → mean 8.0

HARDNESS RESULT

EDTA and metal combine 1 : 1, so 1 mol of EDTA ≡ 1 mol of $CaCO_3$ ≡ 100 g.

Total: moles EDTA = 0.0100 × 0.0125 = 1.25×10^{-4} mol → $CaCO_3$ equivalent = 1.25×10^{-4} × 100 g = 12.5 mg in 50.0 mL.

Total hardness = 12.5 mg × (1000⁄50.0) = **250 mg/L as $CaCO_3$**.

In one line, for a titre V mL of M-molar EDTA with a 50.0 mL sample: hardness = V × M × 100,000⁄50.0.

Permanent: 8.0 × 0.0100 × 100,000⁄50.0 = **160 mg/L**. **Temporary:** 250 − 160 = **90 mg/L**.

On the commonly used scale (< 75 soft; 75–150 moderately hard; 150–300 hard; > 300 very hard) this supply is **hard** — well outside the ~50 mg/L target for wet processing, so it must be softened or the baths sequestered before use.

4. Precautions

- Add the buffer **before** the indicator, and titrate promptly afterwards: at pH 10, standing in air lets the sample absorb CO_2 and lets $Mg(OH)_2$ begin to separate, both of which blur the end point.
- Approach the end point drop by drop with vigorous swirling — the EDTA–indicator exchange is not instant, and a hasty titre overshoots into a blue that arrives one drop too late.
- The boiled determination fails if the make-up volume is forgotten: after filtering, the filtrate must be restored to the original volume, or the permanent hardness is overstated.
- Iron and manganese in quantity block the indicator (the red never releases). In routine work they are masked (e.g. with hydroxylamine hydrochloride or triethanolamine) before titration; note their presence when the end point drags.
- Rinse the burette with EDTA solution and the pipette with the sample; rinse the flask with distilled water only. Read the meniscus without parallax, and reject any titre from a flask that shows a lingering reddish tinge.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Name the ions that cause hardness, and the salts behind temporary and permanent hardness respectively.
2. Why is hardness reported “as $CaCO_3$” when the water may contain almost no $CaCO_3$?

3. State the combining ratio of EDTA with Ca^{2+} and Mg^{2+}, and the colour change and pH of the EBT end point.
4. A 25.0 mL sample needs 8.0 mL of 0.0100 M EDTA. Calculate its hardness in mg/L as $CaCO_3$ and classify it on the scale used in this sheet.
5. Why does boiling fail to cure permanent hardness? Support the answer with one reaction that boiling *does* drive.
6. Match the remedy to the situation: lime–soda, ion exchange, sequestration — which softens the factory supply, and which protects a single dye bath, and why?

Answer key

1. Ca^{2+} and Mg^{2+} (Fe/Mn minor). Temporary: the bicarbonates $Ca(HCO_3)_2$ and $Mg(HCO_3)_2$. Permanent: the chlorides and sulphates — $CaCl_2$, $MgCl_2$, $CaSO_4$, $MgSO_4$.
2. Hardness is a combined effect of several salts; expressing all of it as the equivalent amount of one reference substance — $CaCO_3$, equivalent weight 50 — gives a single comparable number whatever the mixture.
3. 1 : 1 with each metal ion. End point at pH 10 (ammonia buffer): wine red (metal–indicator complex) to pure blue (free indicator) as EDTA strips the last metal from EBT.
4. Moles EDTA = $0.0100 \times 0.0080 = 8.0 \times 10^{-5}$ mol → 8.0 mg $CaCO_3$ in 25.0 mL → $8.0 \times (1000/25.0)$ = **320 mg/L** — very hard (above 300).
5. Permanent salts (chlorides, sulphates) are thermally stable in solution — no gas is expelled and nothing precipitates on boiling. Boiling only destroys bicarbonates: $Ca(HCO_3)_2 \rightarrow CaCO_3\downarrow + CO_2\uparrow + H_2O$, because the CO_2 that held them dissolved is driven off.
6. Ion exchange (or lime–soda for very large volumes) softens the whole supply at the plant inlet — hardness is removed once, for every bath. Sequestration removes nothing; it complexes the Ca^{2+}/Mg^{2+} inside one bath so they cannot act during that process — the tool for a single critical bath or as insurance on partly softened water.
